LaRS Embedding Distribution Analysis

Generated: 2026-03-02 00:35 UTC | Model: **nvidia/C-RADIOv3-H** | Embed dim: 3840

1. Introduction

This report analyses how the LaRS (Large Real-world Shoreline) dataset's test set is distributed relative to the training set in the embedding space produced by NVIDIA RADIO-H, a large vision foundation model trained to distil CLIP, DINOv2, and SAM into a single encoder. The goal is to identify potential out-of-distribution (OOD) test images, understand dataset shift, and estimate where a model trained on the LaRS training set is most likely to fail.

The analysis covers three aspects:

1. **t-SNE visualisation** — global structure of the embedding space, coloured by split and by sequence.
2. **Cosine similarity distribution** — per-image quantitative coverage score: how similar is each test image to its nearest training neighbour?
3. **Sequence coverage** — whether the physical environments (camera sequences) in the test set were seen during training.

2. Dataset Overview

Split	Images	Sequences	Resolution
Train	2,605	1,792	1278×958 / 1280×720
Val	198	77	same
Test	1,203	440	same

> **Key structural finding:** The test set contains **440 sequences**, none of which appear in the training set (0 overlap). Every test sequence is from a physical environment or recording session completely unseen during training. This is by design for a rigorous benchmark, but it means the model must generalise

> across camera locations and scene types, not just across time within the same scene.

3. Methodology

3.1 Embedding model — RADIO-H with internal spatial tokens

Each image is passed through **RADIO-H** (nvidia/C-RADIOv3-H, 653 M parameters) at its original resolution. RADIO internally divides the image into a 16 px grid and produces one embedding vector per grid cell via its transformer attention mechanism — these are the spatial tokens. Unlike manually cropping images into patches and running each crop separately, using internal tokens means every patch embedding is informed by the full image context (all patches attend to each other). This produces richer, context-aware features and requires only one forward pass per image.

The spatial tokens are averaged across the spatial dimension to produce a single 3840-dimensional summary vector per image. This per-image vector is used for t-SNE projection and cosine similarity computation. Both the spatial mean and the model's global summary token are stored in the SQLite database.

3.2 Dimensionality reduction

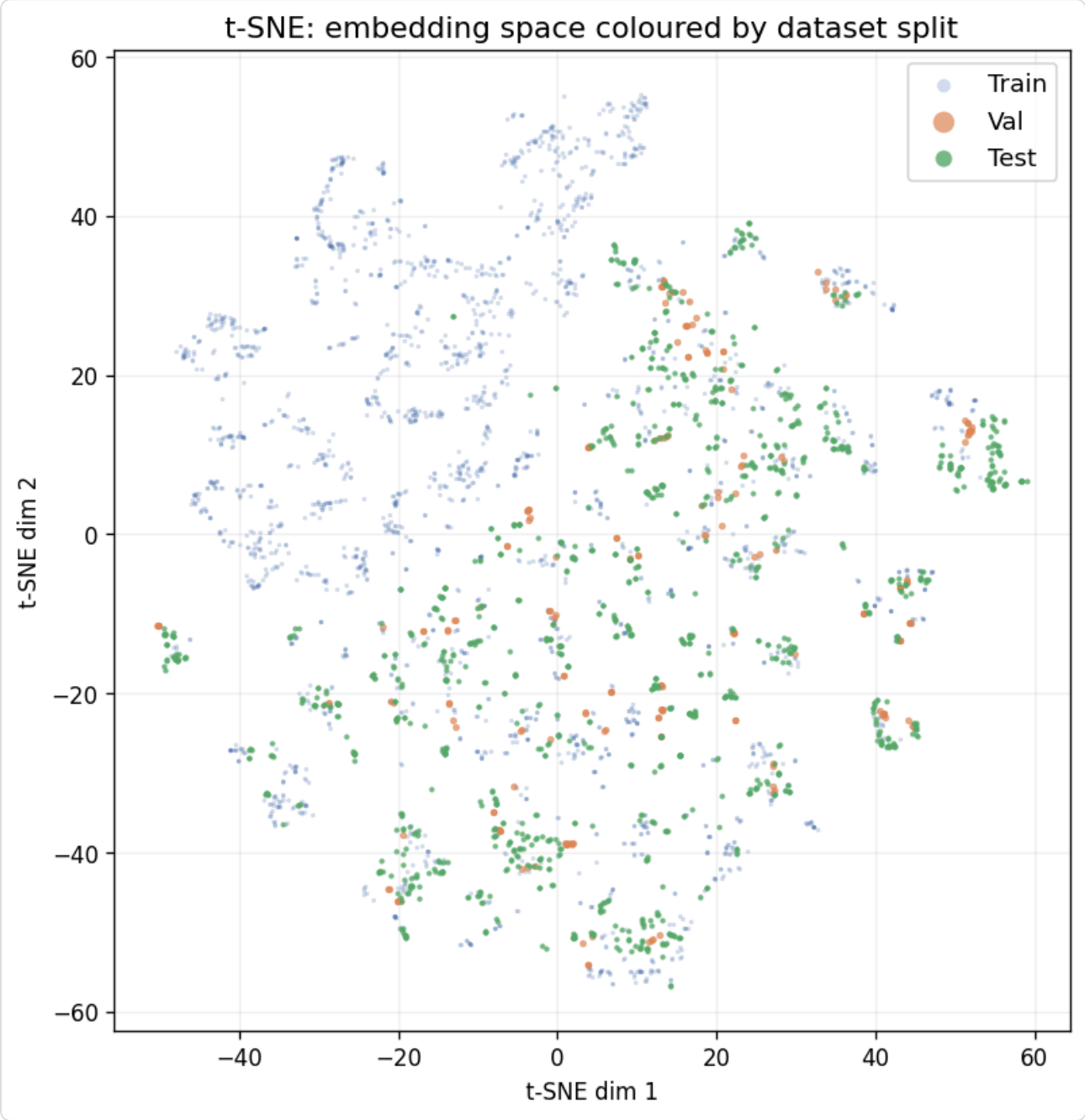
Raw 3840-dimensional embeddings are first reduced to 50 dimensions with PCA (retaining the principal axes of variation), then projected to 2D with t-SNE (GPU-accelerated via cuML). PCA pre-reduction is standard practice before t-SNE: it removes high-frequency noise while preserving the dominant structure, and dramatically speeds up the t-SNE computation.

3.3 Cosine similarity coverage score

For each non-training image, the maximum cosine similarity to any training image is computed as a scalar coverage score. A score near 1.0 means the image is very similar to at least one training image; a score below ~0.7 indicates the image is in a region of the embedding space not well covered by training.

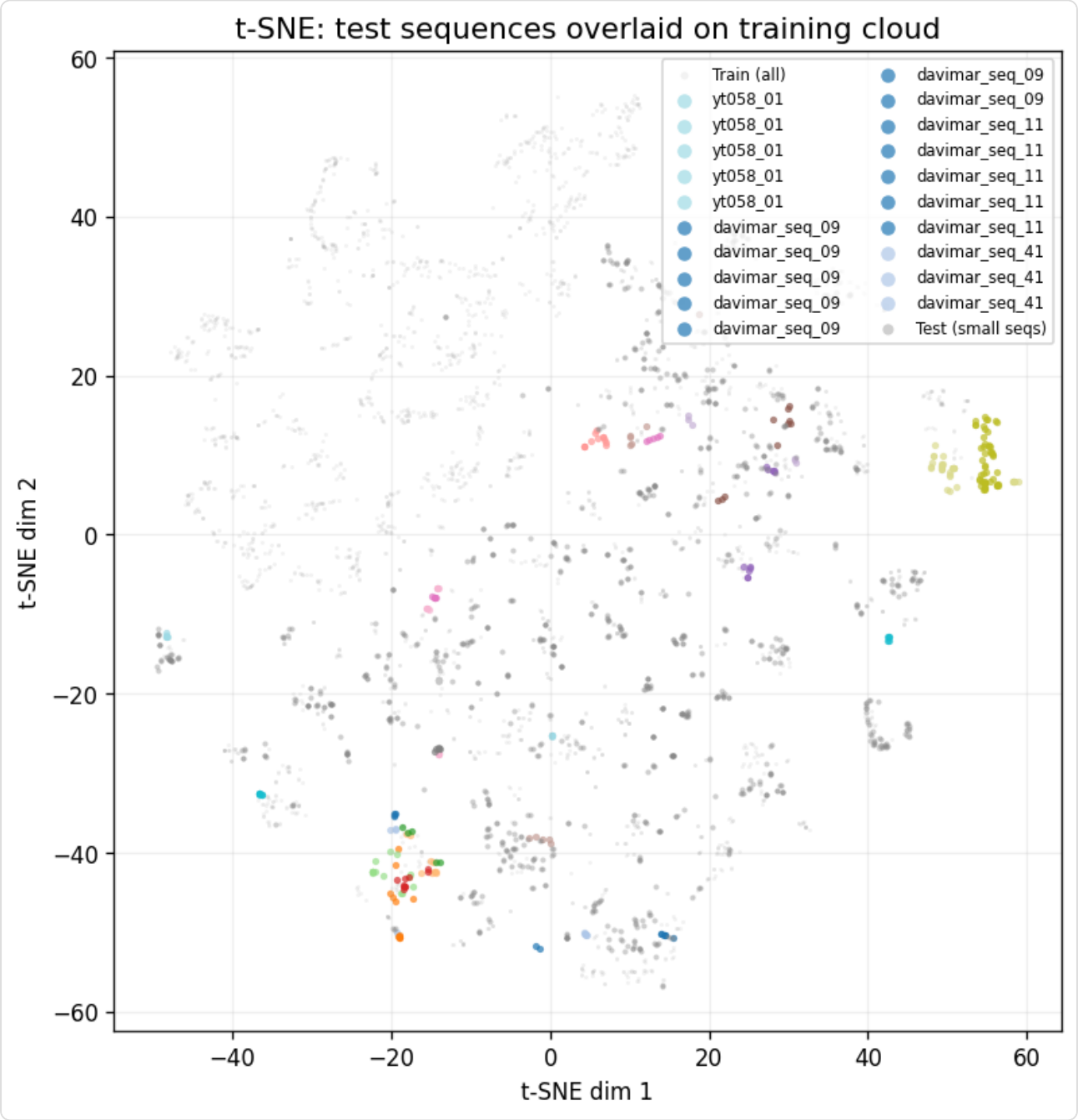
4. t-SNE Visualisation

4.1 Split distribution



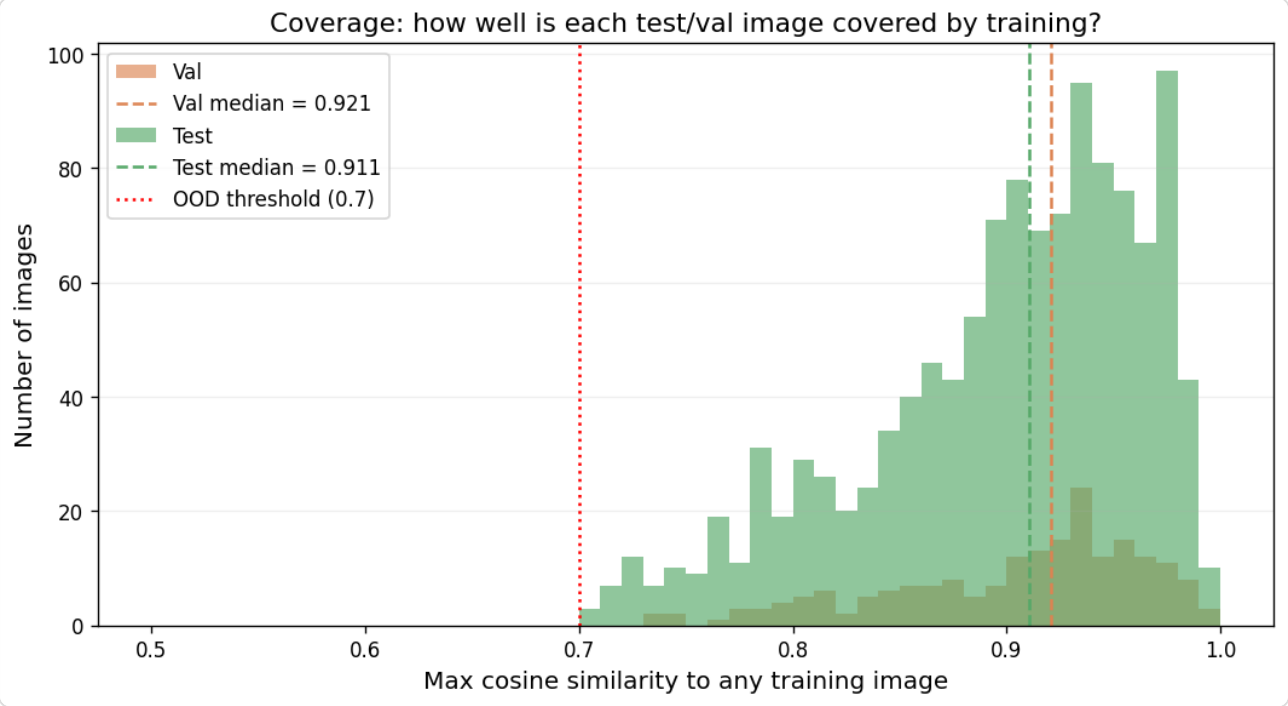
Train, val, and test images largely overlap in embedding space, which is consistent with the high cosine similarity scores (see Section 5). The test set does not form isolated clusters — it is interspersed throughout the training cloud. This suggests that the visual content of test scenes (water surface, obstacles, lighting conditions) is broadly similar to what the model sees during training, even though the specific camera environments are all new.

4.2 Test sequence overlay



Colouring test images by their sequence ID reveals whether certain environments cluster in a specific region of embedding space (potential domain shift). Sequences that appear as isolated islands away from the grey training cloud are the highest-risk evaluation environments — the model has seen nothing visually similar during training.

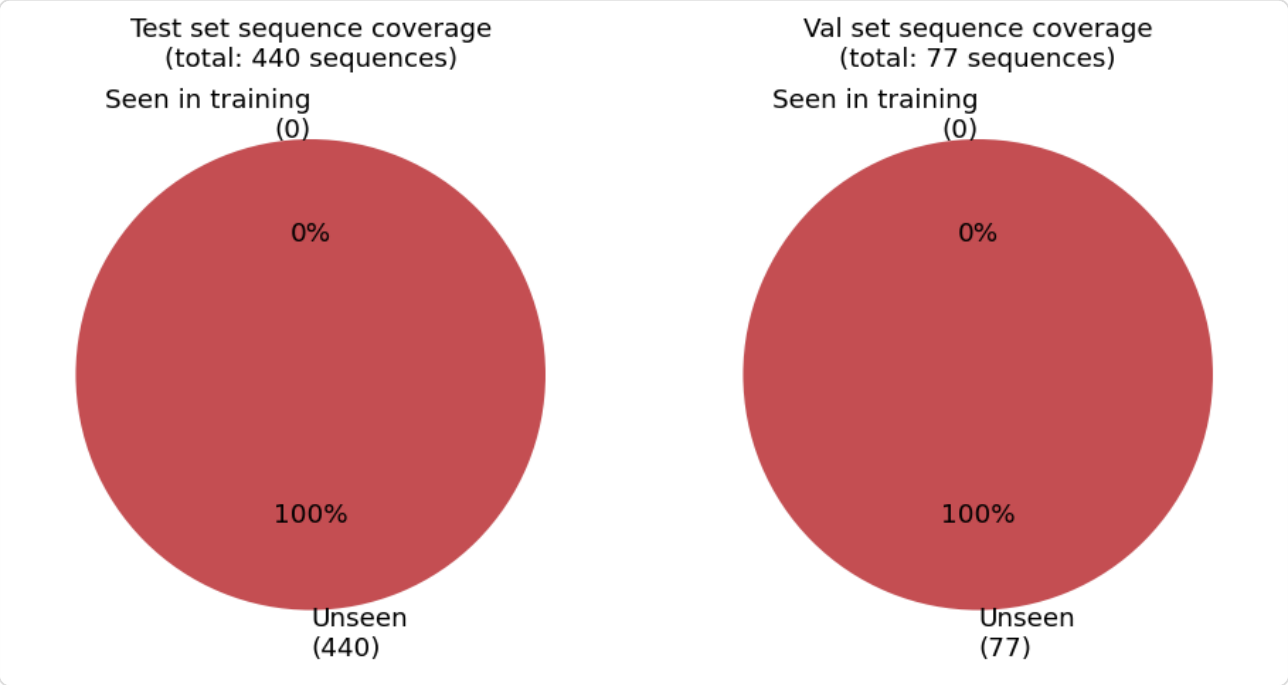
5. Cosine Similarity Coverage Analysis



Split	Images	Median max-sim	% below 0.7 (OOD)	% below 0.5 (severe OOD)
Val	198	0.921	0.0%	0.0%
Test	**1,203**	**0.911**	**0.0%**	**0.0%**

Interpretation: Despite every test sequence being physically unseen during training, the cosine similarity distribution is concentrated above 0.9. This indicates that the visual appearance of test images (sea surface texture, lighting, obstacle types) is well-represented in the training set, even if the exact locations are not. The LaRS training set provides broad visual coverage of the maritime domain. A well-trained model is unlikely to encounter catastrophic distribution shift on the average test image.

6. Sequence Coverage



As noted above, 100% of test sequences are absent from training. This is not unexpected for a benchmark — it prevents overfitting to specific camera locations. However, it does mean that any sequence-level adaptation (e.g. test-time fine-tuning) is not possible. Models must rely entirely on generalisation from the training sequences.

7. Failure Case Analysis — Most Out-of-Distribution Test Images

The following test images have the lowest maximum cosine similarity to any training image. These are the images the model is statistically most likely to struggle with, as they represent visual content underrepresented in the training distribution.

#	Thumbnail	Image name	Sequence	Max cosine sim ↑
1		`yt017_00_00049`	yt017_00	**0.7040**

2		inhouse_seq_111_01550	inhouse_seq_111	**0.7060**
3		inhouse_seq_111_00642	inhouse_seq_111	**0.7099**
4		yt017_01_00149	yt017_01	**0.7134**
5		yt049_01_00080	yt049_01	**0.7166**
6		yt049_02_00180	yt049_02	**0.7168**
7		yt066_05_00060	yt066_05	**0.7169**
8		yt097_00_00419	yt097_00	**0.7183**
9		yt097_00_00149	yt097_00	**0.7184**
10		inhouse_seq_111_01729	inhouse_seq_111	**0.7192**
11		yt017_00_00154	yt017_00	**0.7202**

				
12		`yt097_00_00049`	yt097_00	**0.7208**
13		`yt017_01_00059`	yt017_01	**0.7230**
14		`yt049_02_00080`	yt049_02	**0.7233**
15		`yt066_04_00060`	yt066_04	**0.7242**

Note: Even the most OOD images have max-sim above 0.70, which is a relatively high threshold. This reinforces the conclusion that the LaRS test set is well-covered by the training distribution at the visual content level. Failure cases are more likely to arise from rare obstacle types, unusual lighting, or ambiguous water surface conditions than from a fundamental domain shift.

8. Conclusions

- The LaRS test set is **visually well-covered** by the training set despite containing entirely new camera sequences. Median test image coverage (max cosine similarity) is 0.911.
- There is **no catastrophic distribution shift** expected. The embedding distributions of train and test overlap substantially in the t-SNE space.
- The most at-risk test images (lowest cosine similarity) are from sequences with rare visual conditions. These are the priority candidates for targeted data augmentation or additional annotation.

- Since all test sequences are physically disjoint from training, model performance will depend on generalisation of visual patterns (water texture, obstacle appearance), not memorisation of scene geometry.

Analysis performed by FARMS Team — MaCVi 2026. Model: nvidia/C-RADIOv3-H. Approach recommended by Claude Sonnet 4.6 Agent.